

BALL MILL 5

MAINTENANCE MANUAL

Fictional Industrial Equipment Documentation — Prototype Demonstration

Document No.	BM05-MM-001
Equipment Tag	BM-05
Equipment	Ball Mill 5
Model	FPM-BM5200-2C (Fictional)
Application	Cement Grinding
Revision	Rev. 01
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Status	SAMPLE / NON-OPERATIONAL

IMPORTANT: This is a fictional maintenance manual created for a digital-documentation prototype. All equipment ratings, dimensions, clearances, lubricant specifications, procedures, component references and maintenance intervals are invented. It must not be used to service real machinery.

1. DOCUMENT CONTROL AND MAINTENANCE PHILOSOPHY

Rev.	Date	Description	Prepared	Approval
00	15 Aug 2026	Initial draft	Engineering Systems	Prototype Review
01	01 Sep 2026	Prototype issue	Engineering Systems	Prototype Review

1.1 Purpose

This manual establishes a fictional preventive and corrective maintenance framework for BM-05. It covers inspection, lubrication, mechanical assemblies, drive components, instrumentation checks and planned shutdown activities.

1.2 Maintenance Philosophy

Maintenance is based on a combination of preventive inspection, condition monitoring and planned corrective work. Maintenance frequency should be adjusted in real applications using equipment history, manufacturer requirements, operating severity and condition-monitoring trends.

1.3 Maintenance Principles

- Plan work before opening guarded or rotating equipment.
- Use approved isolation and permit-to-work procedures.
- Record measured values rather than relying only on visual observations.
- Investigate trends and repeated failures rather than repeatedly resetting alarms.
- Replace damaged fasteners, seals and safety devices with approved parts.
- Verify guards and protective devices before returning equipment to service.

1.4 Related Documents

Document	Reference
Operating Manual	BM05-OM-001
Electrical Drawing	BM05-EL-001
Troubleshooting Guide	BM05-TG-001
Inspection Checklist	BM05-CK-001

2. EQUIPMENT OVERVIEW AND MAINTENANCE POINTS

2.1 Principal Assemblies

Assembly	Maintenance Focus
Mill shell	Visual condition, cracks, leakage, shell fasteners and attached components.
Main bearings	Temperature, oil flow, seals, vibration and bearing condition.
Gearbox	Oil condition, temperature, vibration, coupling and gear condition.
Main drive motor	Bearings, cooling, terminals, insulation and vibration.
Intermediate diaphragm	Wear, grate condition, bolts and material passage.
Chamber liners	Wear profile, bolts, cracks and loose sections.
Lubrication unit	Oil level, filters, pump, cooler, pressure and flow switches.
Mill inlet/outlet	Seals, buildup, wear and material flow.

2.2 Maintenance Identification Tags

Tag	Component	Typical Measurement
BRG-05A	Drive-end main bearing	Temperature / vibration
BRG-05B	Non-drive-end main bearing	Temperature / vibration
GB-05	Main gearbox	Oil temperature / vibration
MTR-05	Main motor	Current / temperature / vibration
LUB-05	Central lubrication skid	Pressure / flow / oil level
DIA-05	Intermediate diaphragm	Wear / condition

2.3 Condition Monitoring

Fictional condition-monitoring data include bearing temperature, vibration velocity, oil pressure, oil temperature and motor current. Baselines should be established after commissioning and compared with subsequent readings.

3. SAFETY, ISOLATION AND PREPARATION

SAFETY WARNING: A real ball mill is a high-energy machine. Before maintenance, the responsible team must follow the facility's approved lockout/tagout, electrical isolation, mechanical isolation, confined-space, work-at-height and permit procedures. This sample document does not replace those controls.

3.1 Minimum Preparation

- Review work order and scope.
- Identify all electrical, mechanical, hydraulic, pneumatic and process energy sources.
- Stop and isolate the equipment using the approved site procedure.
- Apply personal locks/tags as required.
- Verify zero-energy state using approved test methods.
- Allow hot components to cool before inspection.
- Barricade the work area and control access.
- Confirm lifting equipment and rated lifting points before moving heavy components.

3.2 Mill Internal Access

Internal access is a high-risk activity. In a real installation it would require a formal risk assessment, isolation verification, confined-space controls where applicable, ventilation, lighting, communication, atmospheric testing where required, standby arrangements and an approved entry permit.

3.3 Tools and Measuring Equipment

Tool	Typical Use
Torque wrench	Controlled tightening of critical fasteners
Dial indicator	Alignment and runout checks
Vibration meter	Condition monitoring
Infrared thermometer	Surface temperature comparison
Feeler gauges	Clearance measurement
Oil sampling kit	Lubricant condition assessment
Calibrated multimeter	Electrical verification by authorized personnel

4. PREVENTIVE MAINTENANCE PROGRAM

Frequency	Maintenance Activity
Each shift	Check leaks, abnormal noise, vibration indication, bearing temperatures and lubrication status.
Daily	Inspect lubrication skid, gearbox, seals, guards and visible fasteners.
Weekly	Review trends; inspect oil filters/indicators and visible drive components.
Monthly	Inspect coupling condition, foundation/anchor areas and instrumentation.
Quarterly	Perform detailed vibration review, oil sampling and selected torque/condition checks.
6 months	Inspect lubrication system, cooler performance, seals and selected mechanical clearances.
Annual shutdown	Detailed inspection of liners, diaphragm, bearings, gearbox, coupling, motor and structural items.

4.1 Preventive Maintenance Checklist

No.	Inspection	Result / Remarks
01	Main bearing temperature trend	_____
02	Main bearing oil flow	_____
03	Gearbox oil level/condition	_____
04	Lubrication filter condition	_____
05	Gearbox vibration	_____
06	Motor vibration/temperature	_____
07	Mill shell/attached components	_____
08	Visible leakage	_____
09	Guards and safety devices	_____
10	Unusual sound or process condition	_____

4.2 Work Order Closeout

Record the equipment tag, work-order number, date, readings before and after maintenance, parts used, defects found, corrective actions, technician and supervisor approval.

5. LUBRICATION SYSTEM AND MAIN BEARINGS

5.1 Fictional Lubrication Arrangement

BM-05 uses a fictional forced-oil lubrication package supplying the main bearings and drive gearbox. The package consists of a reservoir, duty pump, standby pump, filter, oil cooler, pressure switch/transmitter, temperature indication and distribution piping.

Parameter	Fictional Target
Reservoir level	40–80% sight-glass range
Supply pressure	2.0–3.5 bar
Supply temperature	35–50 °C
Filter differential pressure	Below 1.0 bar
Oil cleanliness	Within site-approved target

5.2 Main Bearing Inspection

- Check housing for oil leakage.
- Compare drive-end and non-drive-end temperatures.
- Review vibration trend for increasing amplitude.
- Inspect seals for damage or deterioration.
- Confirm oil flow to each bearing.
- Investigate sudden changes rather than accepting a new reading as normal.

5.3 Lubricant Inspection

Check oil level and appearance. During scheduled maintenance, take a representative sample using clean sampling equipment. Evaluate viscosity, water contamination, particle contamination and wear metals through an appropriate laboratory or approved site process.

5.4 Filter and Cooler

Monitor filter differential pressure and cooler performance. A rapidly increasing differential pressure may indicate contamination or filter loading. Reduced cooling effectiveness may indicate fouling, inadequate cooling medium or control problems.

5.5 Fictional Lubrication Fault Indicators

Observation	Possible Cause	Action
Low pressure	Low level / pump issue / restriction	Investigate before operation.
High temperature	Cooler issue / degraded oil	Check cooler and oil condition.
High filter DP	Loaded filter / contamination	Inspect filter and oil system.

6. MILL LINERS, DIAPHRAGM AND INTERNALS

6.1 Liner Inspection

During a planned shutdown, inspect the first- and second-chamber liners for wear, cracking, deformation, loose fasteners and unusual wear patterns. Measure representative liner thickness at predefined points and compare against the established replacement criterion.

Inspection Point	Method	Record
Liner thickness	Ultrasonic / mechanical measurement as applicable	_____ mm
Liner bolt condition	Visual / torque verification where specified	OK / Defect
Cracks	Visual inspection	Location / size
Loose sections	Visual / approved inspection method	Location
Wear profile	Template / dimensional measurement	_____

6.2 Intermediate Diaphragm

Inspect diaphragm plates, grate openings, lifters, bolts and central components for wear or blockage. Check for distorted or damaged sections that could restrict material transfer between chambers.

6.3 Replacement Planning

Replacement quantities should be determined from inspection findings and approved spare-parts records. Heavy components must be handled with rated lifting equipment and controlled lifting plans.

6.4 Internal Inspection Record

Area	Condition	Action
Chamber 1 liners	_____	_____
Diaphragm	_____	_____
Chamber 2 liners	_____	_____
Shell interior	_____	_____
Mill discharge	_____	_____

Note: The actual acceptable wear limits, bolt torque values and replacement criteria must come from the real equipment manufacturer and approved site engineering standards.

7. DRIVE TRAIN, GEARBOX AND MOTOR

7.1 Main Drive Inspection

Component	Inspection
Main motor	Check vibration, temperature, cooling, terminals and bearing condition.
Flexible coupling	Inspect elements, hubs, guards and alignment indicators.
Gearbox	Check oil level, temperature, vibration, leakage and abnormal noise.
Pinion / drive interface	Inspect lubrication, wear pattern, alignment and fasteners.
Foundation	Inspect anchor areas, grout condition and signs of movement.

7.2 Alignment Checks

Alignment should be verified using an approved measurement method whenever major drive components are disturbed. Record horizontal and vertical offsets, coupling condition and final alignment values in the maintenance record.

Measurement	Before	After
Horizontal offset	_____ mm	_____ mm
Vertical offset	_____ mm	_____ mm
Angular condition	_____	_____
Coupling gap	_____ mm	_____ mm

7.3 Gearbox Oil

Inspect oil level, color, odor and contamination indicators. Review oil analysis results for viscosity change, water, particles and abnormal wear metals. Any sudden deterioration should trigger an engineering assessment.

7.4 Main Motor

Electrical maintenance must be carried out only by authorized personnel. Mechanical checks may include cooling airflow, bearing condition, vibration, mounting condition and visible cleanliness. Electrical tests should follow the motor manufacturer's approved procedures and site electrical standards.

7.5 Fastener Control

Critical fasteners should be tightened using calibrated tools and the approved torque specification for the actual component. Torque values in this fictional manual are intentionally omitted to prevent their being mistaken for real equipment specifications.

8. CORRECTIVE MAINTENANCE AND FAULT INVESTIGATION

8.1 Corrective Maintenance Workflow

Stage	Action
1. Identify	Confirm symptom, alarm, time and operating condition.
2. Isolate	Apply approved isolation and verify safe state.
3. Inspect	Examine the relevant mechanical/electrical/process interfaces.
4. Diagnose	Use drawings, trends, measurements and manufacturer information.
5. Repair	Correct the root cause using approved parts and procedures.
6. Test	Perform controlled post-maintenance checks.
7. Return	Restore guards, remove tools and complete approved restart.
8. Record	Capture failure mode, cause, repair and recommendations.

8.2 Example Failure: Rising Main Bearing Temperature

Check	Possible Finding	Response
Oil flow	Reduced flow	Inspect pump, filter and restriction.
Oil temperature	High supply temperature	Check cooler and cooling medium.
Bearing vibration	Increasing trend	Investigate alignment, lubrication and mechanical condition.
Seal area	Leakage	Inspect seal and oil containment.
Load	Abnormally high load	Check process and mill operating condition.

8.3 Root-Cause Analysis

For repeated or significant failures, document the failure mode, physical evidence, operating conditions, previous maintenance, contributing factors and root cause. Corrective action should address the cause rather than only replacing the failed component.

8.4 Post-Maintenance Verification

- Verify all tools and temporary materials are removed.
- Confirm guards and covers are correctly installed.
- Verify lubrication and auxiliary systems.
- Check instrumentation and alarms.
- Perform controlled trial operation.
- Monitor the repaired component against baseline readings.
- Close the work order with complete records.

9. SHUTDOWN OVERHAUL CHECKLIST AND RECORDS

9.1 Annual Shutdown Scope — Fictional

Area	Recommended Inspection
Mill shell	Visual examination, attached components and structural condition.
Chamber 1	Liner wear, fasteners and internal condition.
Diaphragm	Grates, bolts, wear parts and material passage.
Chamber 2	Liner wear and internal condition.
Main bearings	Seals, lubrication paths, vibration and condition.
Gearbox	Oil analysis, seals, vibration and gear condition.
Motor	Bearings, cooling and authorized electrical inspection.
Coupling	Elements, hubs, guard and alignment.
Lubrication skid	Pump, filters, cooler, hoses/pipes and instruments.
Instrumentation	Sensor condition, wiring and calibration status.

9.2 Maintenance History Record

Date	Work Order	Finding / Work Performed	Technician	Approval
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____

9.3 Spare Parts Categories

Typical fictional categories include liner components, liner fasteners, bearing seals, lubrication filters, pump seals, gearbox oil, coupling elements, instrumentation sensors and general maintenance consumables. Actual part numbers must always be taken from the approved spare-parts catalogue for the real equipment.

9.4 Final Maintenance Release

Item	Status
Maintenance work complete	■
Tools/materials removed	■
Guards restored	■
Isolation cleared under approved procedure	■
Lubrication system ready	■
Post-maintenance inspection complete	■
Work order and readings recorded	■
Authorized release for operation	■

END OF DOCUMENT — BM05-MM-001 Rev. 01

This ten-page manual is fictional and intended only for demonstration of a digital maintenance-documentation platform. It contains no real manufacturer or company proprietary maintenance instructions.